

Assignment - 01

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COURSE CODE: CSA 5113

COURSE NAME: Cryptography & Network Security

1) *ENCRYPT: "DEFEND THE WALL"

FORMULA:

$$C = (P + 3) \bmod 26 \quad C: \text{Caesar Cipher with key} = 3$$

Shift:

D → G E → H F → I G → J H → K I → L J → M K → N L → O M → P N → Q O → R P → S Q → T R → U S → V T → W U → X V → Y W → Z X → A Y → B Z → C

T → W H → K E → H

W → Z A → D L → O L → O

Ciphertext: GHIKHLG ZHR ZDOO

*DECRYPT "KHOOR ZRUOG" with key = 3

FORMULA:

$$P = (C - 3) \bmod 26$$

Shift

K → H H → E O → L O → L R → O

Z → W R → O U → R O → L G → D

*To find the key box SECRET → VHFUHW and decrypt

"WRLV LV DVAH"

Key: S shifted to V. key = 3

Decrypt: THIS IS A TEST.

2) *ENCRYPT "FIRE"

In QWERTY substitution key.

F $\rightarrow$ Y

I $\rightarrow$ O

R $\rightarrow$ K

E $\rightarrow$ T

Ciphertext: YOKT

* To find plaintext

Ceaser cipher of shift - 3.

$$P = (C + 3) \bmod 26$$

QEB $\rightarrow$ THE

NRFZH $\rightarrow$ QUICK.

YOLTR $\rightarrow$ BROWN

UJHHID $\rightarrow$ XMKLGL.

3) * $P = (C - K) \bmod 26$

$$Q - K = G \quad N - E = J \quad X - Y = Z \quad E - K = U \quad P - E = L$$

$\therefore$ Plaintext: GJZULMCWGJCW

* Ciphertext: LXFOPEFRNRH

$$P = (C - K) \bmod 26$$

Key: LEMON

Plaintext: ATTACK AT DAWN

* Plaintext: Attack at Palace

Key: LEMON

$$C = (P + K) \bmod 26$$

Ciphertext: LXFOPEFRDNWEDS

4) Encrypt "HI" $K = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix}$

* Vector Conversion : H=7 I=8

$$C = K \cdot P \text{ mod } 26 = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 7 \\ 8 \end{pmatrix}$$

$$= \begin{pmatrix} 45 \\ 54 \end{pmatrix}$$

$$45 \text{ mod } 26 = 19$$

$$54 \text{ mod } 26 = 2$$

Ciphertext : TC

* Encrypt "OA" to "QC"

$$P = \begin{pmatrix} 14 \\ 0 \end{pmatrix}$$

$$C = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 14 \\ 0 \end{pmatrix} = \begin{pmatrix} 42 \\ 28 \end{pmatrix}$$

$$42 \text{ mod } 26 = 16 \rightarrow Q$$

$$28 \text{ mod } 26 = 2 \rightarrow C$$

Result : QC.

5) *Encrypt "MEETMEATELUNCH" with "ZEBRAS"

$$\begin{array}{r} 6 \ 3 \ 2 \ 4 \ 1 \ 5 \text{ key} \\ \hline Z \ E \ B \ R \ A \ S \end{array}$$

M E E T M E

A F T E L U

N C H x x x

Ciphertext : MZXETHEFCTEXEXMAN

* Decrypt "EANMTEFCUHEIREL" with "BOARD"

Matrix:

2	4	1	5	3	order
B	O	A	R	D	
M	H	E	R	F	
T	E	A	E	C	
E	T	N	L	U	

plaintext: MHERFTEAECETNLU

b) * Encrypt "HELLO" using one-time pad

$$C = (P + K) \bmod 26$$

$$H + X = 30 \bmod 26 = 4 \rightarrow E$$

$$E + M = 16 \bmod 26 = 16 \rightarrow Q$$

$$L + C = 13 \bmod 26 = 13 \rightarrow N$$

$$L + K = 21 \bmod 26 = 21 \rightarrow V$$

$$L + O = 25 \bmod 26 = 25 \rightarrow Z$$

Ciphertext: EQNVZ

* Under ideal conditions, one-time pad achieves perfect secrecy because the key is completely random, as long as the message, and never reused.

* Because the key is entirely uniform and random, a given ciphertext is equally likely to decrypt into any possible plaintext string of that length.

7) "WE ARE DISCOVERED" with "ZE BRA"

5 3 2 4 1
ZE BRA
WE ARE
DISCOVERED

Ciphertext: EODASREIERCEWDV

* Decrypt "EVLNEACDTKESEABQDFQJDEECUXWN" with "TRIAL"

Key order: A I L R T

28 characters is divided by 5 columns gives 5, 6, 5, 6, 6

Plaintext: DEAR LAWYERS WE ARE DISCOVERED

* INFORMATION SECURITY with HACK.

3 1 2 4
H A C K
IN FO
RM AT
IONS
ECUR
ITY X

Ciphertext: NMCOTFANUYIRIEIOTSRX

8) MONO ALPHABETIC SUBSTITUTION:

Ciphertext: UZASOVUOHXMOFVGHADZPEVSGZW
ZOPFPE SXUZETVCG

Frequency:

O $\rightarrow$ 5 Z $\rightarrow$ 5 P $\rightarrow$ 4 U $\rightarrow$ 4 V $\rightarrow$ 3

X $\rightarrow$ 3 E $\rightarrow$ 3 S $\rightarrow$ 3

Partial decryption:

Mapping Z $\rightarrow$ t, V $\rightarrow$ d, O $\rightarrow$ s, U $\rightarrow$ e, Q $\rightarrow$ w, S $\rightarrow$ a

Plaintext: It was discovered yesterday that several informal

9) Monoalphabetic substitution with plaintext

Ciphertext: BJJW QJW YMJW NX F
XNRUOWX

$$P = (C - S) \bmod 26$$

BJJW $\rightarrow$ WEEK

QJW $\rightarrow$ PER

YMJW $\rightarrow$ THER

NX F XNRUOWX $\rightarrow$ IS A SIMPLE S

10) Vigenère Cipher analysis

$$P = (C - \text{"MATH"} \bmod 26)$$

$$Z - M = 25 - 12 = N$$

$$I - A = I$$

$$C - T = J$$

$$V - H = D$$

Plaintext: NITPS...